CAS CS 538. Solutions to Problem Set 6

Due Thursday March 5, 2026 11:59pm

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How to submit: See instructions on Problem Set 1.

Problem 1. (30 points total) Let F be a secure PRF with n-bit strings for keys, inputs, and outputs.

(a) (5 points) Let $F_1(k, x)$ computed as follows: compute $F(k, x)$, append the last bit of x, and output the result. Prove that F_1 is insecure.

Solution. The main idea behind this proof is the fact that the function, F_1 exposes a part of the input in its output. This can be exploited.

Making the Game

Let $\mathcal{A}$ be the adversary that will play the game of distinguinish F_1 from just some rnd function.

- $\mathcal{A}$ chooses any $x \in^R \{0,1\}^n$
- Push x into Oracle, producing $y \in \{0,1\}^{n+1}$
- Now we check if $x[-1] = y[-1]$
  - If YES, $\hat{b} = 1$
  - Else, $\hat{b} = 0$
- Return $\hat{b}$ to challenger.

Analysis

Let $\text{PRFadv}[\mathcal{A}, F_1] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_1 Assume that the challenger chooses $k \in^R \{0,1\}^n$. In this experiment, the oracle is F_1. Then, by definition, the last bit of the output will ALWAYS be the last bit of x. Therefore $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that every bit of y is uniformly distributed and random to x's bits. Thus, the chance of $\hat{b} = 1$ is $\frac{1}{2}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_1] = |1 - \frac{1}{2}| = \frac{1}{2}$$

This is non-negligible, thus meaning that F_1 is insecure. $\square$

(b) (5 points) Let $F_2(k, x)$ computed as follows: compute $F(k, x)$, append the last bit of k, and output the result. Prove that F_2 is insecure.

Solution. This issue now is that the key is exposed in the output, which can be exploited.

Game

$\mathcal{A}$ is playing to distinguish F_2 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2 \in^R \{0,1\}^n$
- Push x_1, x_2 to Oracle to get $y_1, y_2 \in \{0,1\}^n$
- Now, check if $y_1[-1] == y_2[-1]$. This will set $\hat{b}$
  - YES $\rightarrow 1$
  - ELSE $\rightarrow 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let $\text{PRFadv}[\mathcal{A}, F_2] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k \in^R \{0,1\}^n$. Denote k_{last} to be the last bit of k. In this experiment, the oracle is F_2. Then, by definition, the last bit of outputs y_i will ALWAYS equal to k_{last}. Therefore $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that every bit of y_i is uniformly distributed and random to k's bits. Thus, the chance of $\hat{b} = 1$ is $\frac{1}{2}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_2] = |1 - \frac{1}{2}| = \frac{1}{2}$$

This is non-negligible, thus meaning that F_2 is insecure. $\square$

(c) (10 points) Define F_3 for $2n$-bit keys, $2n$-bit inputs, and $2n$-bit outputs, as follows:

$$F_3((k_1, k_2), (x_1, x_2)) = F(k_1, x_1)||F(k_2, x_2)\,,$$

where $||$ denotes concatenation of strings. Prove that F_3 is insecure.

Solution. The main issue is that parts of the output are dependent on PARTS of the input. I can change just 1 part of the input, and not completely change the output.

Game

$\mathcal{A}$ is playing to distinguish F_3 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2, x_2' \in^R \{0,1\}^n$, where $x_2 \neq x_2'$
- Push (x_1, x_2) to Oracle to get $y = y_1||y_2$, with each $y_i \in \{0,1\}^n$
- Push (x_1, x_2') to Oracle to get $y' = y_1'||y_2'$, with each $y_i' \in \{0,1\}^n$
- Now, check if $y_1 == y_1'$. This will set $\hat{b}$
  - YES $\rightarrow 1$
  - ELSE $\rightarrow 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let PRFadv$[\mathcal{A}, F_3] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k_1, k_2 \in^R \{0,1\}^n$. In this experiment, the oracle is F_3. Note how, in both outputs of y, we're using x_1. That then means that both y_1, y_1' are dependent on x_1. Therefore, it should be the case that $y_1 = y_1'$, and $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This then means that both outputs of both y's bits are uniformly distributed and random to ANY of x_i's bits. Thus, the chance of $\hat{b} = 1$ occuring is $\frac{1}{2^n}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_2] = |1 - \frac{1}{2^n}| \approx 1$$

$\frac{1}{2^n}$ approaches negligiblity as n increases. A negligible minus a non-negligible is non-neglibile (the result is a constant, in simpler terms). Thus, this is non-negligible, meaning that F_3 is insecure. $\square$

(d) (10 points) Define F_4 for $2n$-bit keys, $2n$-bit inputs, and n-bit outputs, as follows:

$$F_4((k_1, k_2), (x_1, x_2)) = F(k_1, x_1) \oplus F(k_2, x_2)\,.$$

Prove that F_4 is insecure.

Solution. The issue, like before, is the fact that sections of the output are dependent on sections of the input, allowing an adversary to modify inputs to find a pattern. It's a bit harder for this case tho, but XORs can be cancelled out.

Game

$\mathcal{A}$ is playing to distinguish F_4 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2, x_1', x_2' \in^R \{0,1\}^n$, where $x_1 \neq x_1', x_2 \neq x_2'$
- Push the following 4 inputs into Oracle to get the following $y_i \in \{0,1\}^n$:
  - $(x_1, x_2) \to$ Oracle $\to y_1$
  - $(x_1, x_2') \to$ Oracle $\to y_2$
  - $(x_1', x_2') \to$ Oracle $\to y_3$
  - $(x_1', x_2) \to$ Oracle $\to y_4$
- Now, check if $y_1 \oplus y_2 \oplus y_3 \oplus y_4 = \{0\}^n$. This will set $\hat{b}$
  - YES $\to 1$
  - ELSE $\to 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let PRFadv$[\mathcal{A}, F_4] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k_1, k_2 \in^R \{0,1\}^n$. In this experiment, the oracle is F_4. Note the number of times each variables x_1, x_2, x_1', x_2' appears in the equation of $y_1 \oplus y_2 \oplus y_3 \oplus y_4$ ($y_i = (x_{\text{smth}}) \oplus (x_{\text{smth}})$). What you'll find is that, each of the following variable I have just listed out will come out EXACTLY TWICE.

Also note how many times k_1, k_2 appears in the same equation. As we're XORing 4 terms, these 2 variables appear an EVEN NUMBER of times. By the properties of XOR, this then means that $y_1 \oplus y_2 \oplus y_3 \oplus y_4$ entirely cancels out to 0. Therefore, $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that each bit of each particular y_i will be uniform and random to one another. However, I am not necessarily saying that each y_i needs to have the same exact bit. $y_1 \oplus y_2 \oplus y_3 = z$ is random, so you could just simlify to check if $z \oplus y_4 = \{0\}^n$. As we're just checking if XORing a rnd value yields all 0s, the chance of this happening is $\frac{1}{2^n}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_4] = |1 - \frac{1}{2^n}| \approx 1$$

$\frac{1}{2^n}$ approaches negligiblity as n increases. A negligible minus a non-negligible is non-neglibile (the result is a constant, in simpler terms). Thus, this is non-negligible, meaning that F_4 is insecure. □

Problem 2. (20 points) There are many PRF designs out there based on different assumptions. Suppose you aren't sure which ones are really secure and want to hedge your bets. In this problem, you will prove that you can take multiple different PRF designs and combine them so that the result is secure as long as at least one them is secure. This type of risk reduction is called a PRF *combiner*. There are also combiners for other cryptographic primitives, like CPA-secure encryption, collision-resistant hashing, etc. In this problem, you will prove that XOR is a good PRF combiner.

Let F_1 and F_2 be functions with n-bit keys, inputs, and outputs. Let $F((k_1, k_2), x) = F_1(k_1, x) \oplus F_2(k_2, x)$. Prove that if either F_1 or F_2 is a secure PRF, then F is a secure PRF. (In fact, it suffices to prove that if F_1 is secure than F is secure, because the proof for F_2 is the same. So just prove it for F_1 to get full credit.)

NB: your proof will, of course, proceed via a reduction. In your reduction, be sure to analyze two cases: when the F_1 adversary is in Experiment 0 of the PRF Attack Game (Game 4.2 in Section 4.4.1), you need to prove that the F adversary is also seeing Experiment 0; when the F_1 adversary is in Experiment 1, you need to prove that the F adversary is also seeing Experiment 1.

Solution. To solve this question, the following lemma is be essentially to proving my point:

Lemma 1 (Random $\oplus$ Fixed)**.** *Let R be a random function, and C be **any** fixed function. Then, the following equation will also be random:*

$$H(x) = R(x) \oplus C(x)$$

Lemma Proof. Imagine R being just a rnd bit (0 or 1), and C was $\{1\}$. Then, $R \oplus C$ outputting 1 only occurs 50% of the time (it is random).

Abstract this to an n-bit function for R, C now. The probability of $R(x) \oplus C(x) = Z$, with Z just being some desired n-bit output, will then also be random. This proves the lemma.

Game

Note: I'll be proving the contrapositive: If F is insecure, then F_1 is also insecure.

Let $\mathcal{B}$ is the adversary playing against F_1, and $\mathcal{A}$ be the game playing against F itself. I'll only really describe $\mathcal{B}$ in detail, $\mathcal{A}$ will be described more abstractly.

A $\mathcal{A}$ chooses some x

- x gets pushed into $\mathcal{B}$

B x gets pushed into oracle, producing $y = F_1(k_1, x)$

- $\mathcal{B}$ chooses $k_2 = \{0,1\}^n$
- $\mathcal{B}$ calculates $z = y \oplus F_2(k_2, x)$
- $\mathcal{B}$ returns z to $\mathcal{A}$

A, B $\mathcal{A}$ determines value of $\hat{b}$ from z

- A, B returns $\hat{b}$

Analysis

I won't explicity write out the $\Pr$ equation right now. HOWEVER note that experiment 0 denotes the psuedo-rnd experiment, and 1 denotes the truly random experiment for BOTH $\mathcal{A}, \mathcal{B}$.

Game 0 (For $\mathcal{B}$) In this game, the oracle $\mathcal{B}$ uses is F_1. Note how both terms of the value that $\mathcal{B}$ outputs to $\mathcal{A}$ are deterministic, $z = F_1(k_1, x) \oplus F_2(k_2, x)$. Further note how game we have just described is the exact same game as the problem asks of us. This means that the experiment $\mathcal{A}$ is playing then is it's pseudo-rnd experiment as well. As $\mathcal{B}$'s output is dependent on $\mathcal{A}$, then:

$$\Pr[W_0^{\mathcal{B}}] = \Pr[W_0^{\mathcal{A}}]$$

Game 1 ($\mathcal{B}$) In this game, the orcle $\mathcal{B}$ now uses is just some rnd func. Now note the terms that $\mathcal{B}$ outputs to $\mathcal{A}$, $z = R(x) \oplus F_2(k_2, x)$. This is just 1, which means that z is a random value. That then means that the game $\mathcal{A}$ will play will then be its rnd experiment. Thus:

$$\Pr[W_1^{\mathcal{B}}] = \Pr[W_1^{\mathcal{A}}]$$

Adv

The advantage for game $\mathcal{B}$ can be calculated as the following:

$$\mathsf{PRFadv}[\mathcal{B}, F_1] = |\Pr[W_0^{\mathcal{B}}] - \Pr[W_1^{\mathcal{B}}]|$$

By our analysis, our probabilities are the exact same as $\mathcal{A}$'s probabilties:

$$|\Pr[W_0^{\mathcal{B}}] - \Pr[W_1^{\mathcal{B}}]| = |\Pr[W_0^{\mathcal{A}}] - \Pr[W_1^{\mathcal{A}}]| = \epsilon$$

As ϵ is assumed to be non-negligble (from our assumption), then this proves that F_1 is insecure, proving our contrapositive statement, and thus proving the OG statement. $\square$

Problem 3. (20 points, at 10 each)

(a) Let $F : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with $\mathcal{K} = \mathcal{X} = \mathcal{Y} = \{0,1\}^\lambda$ be a secure PRF. Define $F^1(k, x)$ is follows: if $k = 0^\lambda$, output 0^λ; else output $F(k, x)$. Prove that F^1 is secure. Hint: use the Difference Lemma (Theorem 4.7 in

Section 4.4.3).

Solution. The hint does heavy lifting here. Basically this is a matter of showing how the probability of getting the 'bad event' is negligble, which bounds to the overall probability of the PRF.

Lemma 2 (Difference)**.** *Already learned it in class, so I'll just put the equation it states down here:*

$$|\Pr[W_0] - \Pr[W_1]| \leqslant \Pr[Z]$$

Game

Define $\mathcal{A}$ as some abstract game that tries to distinguish F^1 from a truly rnd func.

There are 3 total branches $\mathcal{A}$ could take, and it regards the oracle it uses: 2 from the psuedo-rnd experiment, and 1 for the truly random:

Game 0 F^1 is run

Game 1 F is run

Game 2 A rnd func is run

Analysis

Denote $\mathsf{PRFadv}[\mathcal{A}, F] = |\Pr[W_0] - \Pr[W_1] - \Pr[W_2]|$ as the magnitude of probability difference of either game 0, 1, or 2 of occuring. By the triangle inequality, this entire equation is bounded to the following expression:

$$\mathsf{PRFadv}[\mathcal{A}, F] \leqslant |\Pr[W_0] - \Pr[W_1]| + |\Pr[W_1] - \Pr[W_2]|$$

We can now analyze these 2 terms.

Game 0 → Game 1 Note how the literal difference in what $\mathcal{A}$ does between Game 0 and Game 1 is the exact same, EXCEPT when $k = 0^\lambda$. Denote the event of this occuring as event $\mathbb{Z}$. Then, by 2, the expression $|\Pr[W_0] - \Pr[W_1]|$ is bounded to $\Pr[\mathbb{Z}]$. Event $\mathbb{Z}$ only occurs when 1 particular element in $\mathcal{K}$ occurs (that is, 0^λ). So then:

$$|\Pr[W_0] - \Pr[W_1]| \leqslant \Pr[\mathbb{Z}] \leqslant \frac{1}{2^\lambda}$$

Game 1 → Game 2 Game 1 is assumed to be secure. Game 2 uses a rnd oracle, so the output is uniform and rnd to any possible check $\mathcal{A}$ could do. These are just 2 negligibles:

$$|\Pr[W_1] - \Pr[W_2]| <= \text{negligible}(\lambda)$$

Adv

So then, the overall advantage of this question becomes bounded to the following:

$$\text{PRFadv}[\mathcal{A}, F] \leqslant |\frac{1}{2^\lambda}| + |\text{negligible}(\lambda)|$$

Subtracting 2 negligibles still results in negligible. Thus, F is secure. $\square$

(b) (10 points) Use the previous part to prove the following. Let $H : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with $\mathcal{K} = \mathcal{X} = \mathcal{Y} = \{0,1\}^\lambda$ be a secure PRF. Define $H^1(k', x')$ as follows: set $k = x'$ and $x = k'$; output $H(k, x)$ (that is, simply swap the key and the output). Prove that H^1 is not necessarily a secure PRF.

Solution. The issue here becomes two-fold. Firstly, we're using the input (smth the Adversary CAN control) as a conditional within the PRF. Next is the weakness to the key, k. That weakness becomes visible here.

Game

Let $\mathcal{A}$ be the adversary distinguishing H' from a rnd func.

- $\mathcal{A}$ chooses $x' \in \{0\}^\lambda$
- Push $(k', x') \to$ Oracle $\to y \in \{0,1\}^\lambda$
- Check now if $y == x'$. This is setting $\hat{b}$
  - YES $\to$ 1
  - NO $\to$ 0
- Return $\hat{b}$ from $\mathcal{A}$

Analysis

Let PRFadv$[\mathcal{A}, H'] = |\Pr[W_1] - \Pr[W_2]|$ be the magnitude of difference between events $0, 1$ from outputting 1 from $\mathcal{A}$.

Game 0 In this game, the oracle used is $H'(k', x') \to H(x', k')$. Note how the key here is now the input, x', which the adversary set as $\{0\}^\lambda$. By definition of $H(k, x)$, this then means that y must also then be 0^λ. Thus $\Pr[W_0] = 1$

Game 1 In this game, a rnd func is used by oracle. As the outputs' bits are independent to the input, the probability is just $\frac{1}{2^\lambda}$.

Adv

Thus the advantage is the following:

$$\text{PRFadv}[\mathcal{A}, H'] = |1 - \frac{1}{2^\lambda}| \approx 1$$

A negligible minus a non-negligible is non-negligible (the difference is a const). Also, $\frac{1}{2^\lambda}$ approaches negligible as λ increases. Therefore, this proves that H' is insecure. $\square$

Problem 4. (30 points, at 15 each) Problem 5.3 from Section 5.8 of the Boneh + Shoup textbook. Specifically,

(a) Assume that $\mathcal{E}$ is CPA-secure and prove that $\mathcal{E}$ is Alt-CPA-secure.

(b) Assume that $\mathcal{E}$ is Alt-CPA-secure and prove that $\mathcal{E}$ is CPA-secure.

Hint 1: Do these parts one a time, to avoid getting lost. Part (b) requires a hybrid argument with many hybrid points.

Hint 2: The textbook presents the Alt-CPA definition as a single randomized experiment, with the bit b chosen at random by the challenger, and the adversary is trying to guess b by outputting a matching $\hat{b}$. This definition style is called "bit-guessing"; define

$$\text{Alt-CPAadv}^*(\mathcal{A}, \mathcal{E}) = |\Pr[b = \hat{b}] - 1/2| \,.$$

This definition is equivalent to a two-experiment definition that we are more familiar with: one for $b = 0$, and the other for $b = 1$. In this view, there is no random choice of b, and we are not measuring whether $b = \hat{b}$; rather, we measure the probability difference between $p_0 = \Pr[\hat{b} = 1]$ in Experiment 0 and $p_1 = \Pr[\hat{b} = 1]$ in Experiment 1; define

$$\text{Alt-CPAadv}(\mathcal{A}, \mathcal{E}) = |p_1 - p_0| \,.$$

The two definitions are equivalent; this equivalence is shown in Section 2.2.5 in the textbook (in the context of semantic security, but it's the same for all other such definitions). It is easier to use CPAadv* / Alt-CPAadv* for part (a), and CPAadv / Alt-CPAadv for part (b).

Solution.

A, CPAadv

The solution to part (a) of this question simply involves creating a reduction between Alt-CPAadv, CPAadv. In this case, $\mathcal{A}$ = the Alt-CPA fighter. The goal is to make $\mathcal{B}$, the CPA fighter, perfectly simulate $\mathcal{A}$.

We try to solve the contrapositive: If $\mathcal{E}$ isn't Alt-CPA secure, then $\mathcal{E}$ isn't CPA secure.

Game

Let $\mathcal{A}$ be the adversary fighting the Alt-CPA adversary. $\mathcal{B}$ is the adversary fighting the CPA adversary. $\mathcal{A}$ runs within $\mathcal{B}$.

Firstly, I'll describe $\mathcal{A}$:

- IF $\mathcal{A}$ decides to run an enc query:
  - $\mathcal{A}$ chooses $m \in \mathcal{M}$
  - Pushes $m \to$ Oracle
  - Gets $c \in \mathsf{Enc}(k, m)$
  - This whole loop can be run multiple times
- IF $\mathcal{A}$ decides to run a test query:
  - $\mathcal{A}$ chooses $m_1, m_2 \in \mathcal{M}$
  - Pushes $(m_1, m_2) \to$ Oracle
  - Gets $c \in \mathsf{Enc}(k, m_b)$
  - Can only be run once
- Determines and returns $\hat{b}$

Note that, for $\mathcal{A}$, the oracle will ALWAYS be $\mathcal{B}$ here. Now, for the $\mathcal{B}$ game:

- Gets query from challenger.
- IF query $= m$ (a single var):
  - Sends $(m, m) \to$ oracle
  - Gets $\mathsf{Enc}(k, m)$
  - Sends $\mathsf{Enc}(k, m) \to$ challenger
- IF query $= (m_1, m_2)$ (2-var tuple):
  - Sends $(m_1, m_2) \to$ oracle
  - Gets $\mathsf{Enc}(k, m_b)$
  - Sends $\mathsf{Enc}(k, m_b) \to$ challenger

Note, oracle for $\mathcal{B}$ is the CPA oracle, and challenger is $\mathcal{A}$. Once $\mathcal{A}$ determines and returns $\hat{b}$, $\mathcal{B}$ returns the same value.

Analysis

Let $\text{CPAadv}[\mathcal{B}, \mathcal{E}] = |\Pr[W_0^{\mathcal{B}}] - \Pr[W_1^{\mathcal{B}}]|$ be the magnitude of difference of events $0, 1$ occuring within $\mathcal{B}$.

Also, let $\text{Alt-CPAadv}[\mathcal{A}, \mathcal{E}] = |\Pr[W_0^{\mathcal{A}}] - \Pr[W_1^{\mathcal{A}}] = 2 \times \text{Alt-CPAadv}^*[\mathcal{A}, \mathcal{E}]$ be the same for $\mathcal{A}$. As Alt-CPAadv^* is the bit-guessing advantage, the distinguishing advantage is just half of that. For simplicity, let our inital assumption of that non-negligble advantage, ϵ, be the advatange of Alt-CPAadv, not Alt-CPAadv^*.

Game 0: CPA Oracle encrypts left In the game, the oracle $\mathcal{B}$ interacts with encrypts m_0. Consider the two cases of queries $\mathcal{B}$ could get:

m If $\mathcal{A}$ does an enc test, then $\mathcal{B}$ gets and sends $\mathsf{Enc}(m)$. Note how there's no difference between the bits from $\mathcal{B}$'s input, (m, m). Thus, what $\mathcal{A}$ gets from $\mathcal{B}$ is just the same as what $\mathcal{B}$ got. This is just game $\mathcal{A}$'s enc query.

(m, m) In this case, now $\mathcal{B}$ gets $\mathsf{Enc}(m_0)$. Note how $\mathcal{B}$ sends this output as input to $\mathcal{A}$. This is just game $\mathcal{A}$'s test query.

Thus, the probability distribution of this event will be the same as what would also occur for game $\mathcal{A}$: $\Pr[W_0^{\mathcal{B}}] = \Pr[W_0^{\mathcal{A}}]$

Game 1: CPA Oracle encrypts right In the game, the oracle $\mathcal{B}$ interacts with encrypts m_1. Consider the two cases of queries $\mathcal{B}$ could get:

m If $\mathcal{A}$ does an enc test, then $\mathcal{B}$ gets and sends $\mathsf{Enc}(m)$. Like before, there's no difference between the outputs of m, m, they're the same. Thus, this is just game as $\mathcal{A}$'s enc query.

(m, m) In this case, now $\mathcal{B}$ gets $\mathsf{Enc}(m_1)$. Note how $\mathcal{B}$ sends this output as input to $\mathcal{A}$. This is just game $\mathcal{A}$'s test query.

Same conclusion as before, just for Game 1 now: $\Pr[W_1^{\mathcal{B}}] = \Pr[W_1^{\mathcal{A}}]$

Adv

Thus, the advantage for the CPA game is as follows:

$$\begin{aligned}
\mathsf{CPAadv} &= |\Pr[W_0^{\mathcal{B}}] - \Pr[W_1^{\mathcal{B}}]| \\
&= |\Pr[W_0^{\mathcal{A}}] - \Pr[W_1^{\mathcal{A}}]| \\
&= \mathsf{Alt\text{-}CPAadv}[\mathcal{A}, \mathcal{E}] \\
&= \epsilon
\end{aligned}$$

Our assumption assumed ϵ is non-negligble. Thus, CPAadv is non-negligible.

This proves the contrapositive, and thus the OG statement. $\square$

B, Alt-CPAadv

This proof requires doing a bunch of hybrid arguments. The reason for this is because of what this question really is asking versus the previous one. The previosu question was proving the equivalence of 2 different definitions. But here, I'm forced to now check if Alt-CPAadv's mechanism of single-query is just as secure as the standard CPAadv multi-query test.

I will prove the contrapositive: If $\mathcal{E}$ is **not** CPA-secure, then $\mathcal{E}$ is **not** Alt-CPA-secure.

Hybrid Games

Let $\mathcal{A}$ be the adversary, that can make Q test queries, and is assumed to have non-negligible advantage ϵ against $\mathcal{E}$. Now let $\mathcal{B}$ be the adversary, that can only make 1 query, that uses $\mathcal{A}$ to break Alt-CPA security.

Also, let $Q+1$ be defined as the hybrid games $H_0, H_1, \ldots, H_Q$. It is polynomially bounded:

- **Hybrid H_0:** All Q test queries use $b = 0$ (encrypt $m_0^{(i)}$ for all i)
- **Hybrid H_1:** First query uses $b = 1$, remaining $Q-1$ queries use $b = 0$
- **Hybrid H_2:** First 2 queries use $b = 1$, remaining $Q-2$ queries use $b = 0$
- $\ldots$
- **Hybrid H_Q:** All Q test queries use $b = 1$ (encrypt $m_1^{(i)}$ for all i)

Note how H_0 corresponds to Experiment 0 of the CPA game, and how H_Q corresponds to Experiment 1 of the CPA game.

Adversary

$\mathcal{B}$ is now playing against the Alt-CPA game, and works as the CPA game's oracle within $\mathcal{A}$:

- $\mathcal{B}$ chooses a random value $\omega \in \{1, \ldots, Q\}$
- When $\mathcal{A}$ makes its i-th test query $(m_0^{(i)}, m_1^{(i)})$:
  - If $i < \omega$: $\mathcal{B}$ encrypts $m_1^{(i)}$ using its oracle via an enc query
  - If $i > \omega$: $\mathcal{B}$ will encrypt $m_0^{(i)}$ with oracle via an enc query
  - If $i = \omega$: $\mathcal{B}$ will finally send $(m_0^{(i)}, m_1^{(i)})$ to oracle as a test query, get c^*, and send that to $\mathcal{A}$
- $\mathcal{B}$ returns whatever $\mathcal{A}$ outputs as $\hat{b}$

Analysis

Let W_i be the event that $\mathcal{A}$ outputs 1 in the (hybrid) game H_i.

Case 1: $\mathcal{B}$'s oracle uses $b = 0$ When the Alt-CPA oracle encrypts $m_0^{(\omega)}$, note how $\mathcal{B}$'s game just becomes $H_{\omega-1}$ (as the first $\omega - 1$ queries use $b = 1$, while the rest use $b = 0$).

Case 2: $\mathcal{B}$'s oracle uses $b = 1$ When the Alt-CPA oracle encrypts $m_1^{(\omega)}$, then this is just H_ω (the first ω queries use $b = 1$, rest use $b = 0$).

Adv

The advantage of $\mathcal{B}$ against Alt-CPA thus is:

$$\text{Alt-CPAadv}[\mathcal{B}, \mathcal{E}] = \left| \Pr[\hat{b} = b] - \frac{1}{2} \right|$$

And since ω is independent and random from $\{1, \ldots, Q\}$:

$$\text{Alt-CPAadv}[\mathcal{B}, \mathcal{E}] \leqslant \frac{1}{Q} \sum_{i=1}^{Q} |\Pr[W_i] - \Pr[W_{i-1}]|$$

By the triangle inequality:

$$\text{CPAadv}[\mathcal{A}, \mathcal{E}] = |\Pr[W_Q] - \Pr[W_0]| \leqslant \sum_{i=1}^{Q} |\Pr[W_i] - \Pr[W_{i-1}]|$$

And therefore:

$$\text{Alt-CPAadv}[\mathcal{B}, \mathcal{E}] \geqslant \frac{1}{Q} \times \text{CPAadv}[\mathcal{A}, \mathcal{E}]$$

Our initial assumption tells us that CPAadv will have some non-negligible value. Since Q is polynomially bounded (furthermore, it's $\frac{1}{Q}$, an inverse polynomial), that also then means that that product will still be non-negligible.

This then proves that Alt-CPAadv is non-negligible, and insecure. This proves the contrapositive, which proves our OG statment. $\square$